

An abstract graphic on the left side of the slide, consisting of a network of light blue lines and small circles, resembling a circuit board or a neural network, extending vertically from the top to the bottom of the frame.

REQUIREMENTS ANALYSIS

CLIENT BRIEF

- The client wants to meet the needs of people who enrol on an intensive English language course at their language school. They want to enable their customers to work independently on additional language training which is fun to use. They want users' progress and usage of the product to be able to be viewed by staff. When users first access the product they will need to be able to select their native language and they will then be shown the words on screen in both their native language and English. All recordings will be in English as that is the language the users are learning. The client wants users to be able to access at least three learning activities to learn words and phrases. They would like the user to be able to select any of the three activities to work on.
- One activity should involve repetition of the words and phrases with suitable images and a recording of the word or phrase spoken in English. They would like a record and play back facility so that users can record themselves saying the word or phrase and compare how they sound with the recorded speaker. Another learning activity could be based on memory. This would use the same words, images, recordings of the spoken words and written words as in the first activity. A third activity could be based on the speed and accuracy of the user in associating the spoken words with the correct image. There needs to be some form of scoring system so that users are motivated to achieve high scores and improve. Sound and animation could be included to provide interest and excitement and to celebrate the completion of each section.

CLIENT BRIEF CONTINUED

The words and associated images, text and recorded words must remain consistent throughout. The client initially wants to see what the application looks like and how well it performs for one language (English) and one subject area (snacks and drinks). If this proves successful they would be interested in having the application developed to include more subjects and languages. These are the words and phrases connected with snacks and drinks that the client wants to be used. Initially two other languages have been provided with English so that users can select a native language to use with the product.

English	Italian	Polish
Mineral water	Acqua minerale	Woda mineralna
A black coffee	Un caffè nero	Czarna kawa
Tea with lemon	Tè con limone	Herbata z cytryna
Sugar	Zucchero	Cukier
Milk	Latte	Mleko
Salt	Sale	Sól
Pepper	Pepe	Pieprz
A table for two please	Un tavolo per due per favour	Stolik dla dwojga prosze
Here is the menu	Ecco il menu	Oto menu
Do you want to order?	Volete ordinaire?	Chcesz zamówić?

INTERVIEW

What is your position at the school?

Head teacher

Are you actively involved with the student's education?

Yes

What are the current electronic parts to the current system?

Register, assessment data storage

Roughly how many people would use the system (including staff and students)?

Spread out between 5 groups min 100 5 staff + couple of support staff

What is the available budget to develop this system?

Mid range on a budget

Do you have the necessary strength server to host the system?

No

Do you have anyone in house who would be able and willing to support the development or maintenance of this system e.g., it staff and language teachers?

Yes but reduces budget

What operating system/s will the devices use ?

Windows, android

How should the new system interact with current electronic parts ?

Monitor progress long term as time passes , ability to interact with users , provide feedback on areas they excel in and struggle with

What Is good about the current system?

Tutors multilingual, can help in many languages, take in diverse client, can communicate in native languages, links well with local demographic

What are problems with the current system are?

Engagement and speed of advancement for example students who attend for jobs so low engagement as they see it as just something necessary not something they want to do, students are not all on similar levels so the less advanced ones often get left behind

What do you absolutely expect the new system to do ?

At least 3 activities, optional on which, **gamification** to make them more addictive so they are **more likely to keep coming back**. The app should help the students learn faster and at own standards

What are the things you would like to have from the new system that aren't necessary?

Feedback system, levels of activity, more support, tips, key words, moving up , extrapolate many reports, push notifications

INTERVIEW CONTINUED

How do you wish the systems to interact e.g., new one strictly outside of school time ?

Should not be used at all during school time as it is meant to be a more personalized form of learning, it will most likely will only ever be mentioned for introduction and encouraging its use

how would you like the points system to work e.g. would you reward quantity more or quality, would the amount stay the same if you repeat a module should speed factor in to how many points received?

Should be a focus on quantity of completed tasks for points to encourage repetitive use rather than quality so the amount of points gained by completing the activity perfectly or barely should be mostly the same, it would be preferable for the students to use the app to support their learning as much as possible

Considering the practise task not having an easily verifiable goal how would you like the point system to be implemented

For the practise task in order to earn the points a submission system will have to be created so the teachers can give points when pronunciations are accurate, in the future this would hopefully progress to vocal recognition so points can be given automatically

What would be tracked and viewed by staff (the number of hours overall, average hours per day, highest score at a task, improvement at a task, current number of points)?

Definitely : Overall time spent, Time spent per activity, hopefully: time when the app is used , date of use , setting reminders

PROJECT REQUIREMENTS

Minimum of 3 options of activities for each module available.(core)

- So they can choose which activity based on what they want to practise and or how they enjoy learning the most

Consistent words images and recordings used in all tasks of a module.(core)

- It is more comfortable to use if images and recordings are consistent between the tasks as several images to represent the same thing can disorienting

Point system.(core)

- Necessary for adding incentive to completing activities and improving as it helps them feel like they are actually making progress towards somethings points will be given after each completed activity

All recordings in English.(core)

- Because they are learning English all the recordings would be in English unless the app would be updated to learn other languages

Native language selection(core), words on screen written in both English and native language.(optional)

- This is required as the school has a wide diversity of people who come to learn therefore they need the app to be comprehensible to them or they won't understand and thus won't use it

Progress and usage to be viewable by admins.(optional)

- This will be used to evaluate the students progress(activity results over time),students points ,how much the app is being used, and when it is being used to be able to better understand how to improve the app or the students learning
- For one task need a record, playback and submission system. (optional)
- Task will be used for students to help them improve their pronunciation of the words but currently no way to earn points with the task and no way to know if their pronunciation is improving so this will solve those problems

Must be fun to use and engaging.(N/A [check notes])

- So that the clients are less likely to get bored and more likely to continually use the program to support their learning as there is currently an engagement problem in class from those learning out of necessity

LIMITATIONS AND CONSTRAINTS

- Constraints:
 - Hardware constraints – needs to be built to be layout differently on pc and mobile
 - Development constraints – some things may not be possible do depending on which programming language you use
- Limitations:
 - Scope – it may be more difficult in the future as more knowledge is required the more information is added including languages and module information in those languages
 - I will likely not develop the admin data or the submission system due to these potentially involving interaction between at least 2 devices with the app installed but I may partially if the client sees it as a necessary part of the prototype